

Component modeling and updating method of integrated energy systems based on knowledge distillation

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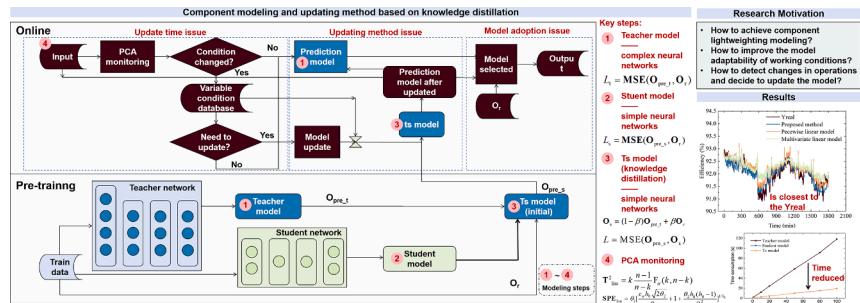
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HIGHLIGHTS

- A component modeling and updating method for IES is proposed.
- Knowledge distillation is utilized for real-time updates of complex models.
- A model update trigger method based on PCA is established.
- The effects of model updates on IES scheduling are analyzed.
- Adaptability of data models to variable conditions is enhanced.

GRAPHICAL ABSTRACT



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ABSTRACT

Amid the backdrop of carbon neutrality, traditional energy production is transitioning towards integrated energy systems (IES), where model-based scheduling is key in scenarios with multiple uncertainties on both supply and demand sides. The development of artificial intelligence algorithms, has resolved issues related to model accuracy. However, under conditions of high proportion renewable energy integration, component load adjustments require increased flexibility, so the mathematical model of the component must adapt to constantly changing operating conditions. Therefore, the identification of operating condition changes and rapid model updating are pressing issues. This study proposes a modeling and updating method for IES components based on knowledge distillation. The core of this modeling method is the light weighting of the model, which is achieved through a knowledge distillation method, using a teacher-student mode to compress complex neural network models. The triggering of model updates is achieved through principal component analysis. The study also analyzes the impact of model errors caused by delayed model updates on the overall scheduling of IES. Case studies are conducted on critical components in IES, including coal-fired boilers and turbines. The results show that the time consumption for model updating is reduced by 76.67 % using the proposed method. Under changing conditions, compared with two traditional models, the average deviation of this method is reduced by 12.61 % and 3.49 %, respectively, thereby improving the model's adaptability. The necessity of updating the component model is

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further analyzed, as a 1.00 % mean squared error in the component model may lead to a power deviation of 0.075 MW. This method provides real-time, adaptable support for IES data modeling and updates.

Nomenclature

Acronyms

MPC	model predictive control
PCA	principal component analysis
DRL	deep reinforcement learning
MSE	mean squared error
IES	integrated energy systems

Symbols

X_0	the original data set with n observations and m variables
X_M	the mean of the X_0
X_V	the variance of the X_0
X	the normalized input data
Z	the covariance matrix
λ	the eigenvalues
$\bar{v}$	the eigenvectors
k	features
d	accumulated contribution limit
P	the feature matrix
T	the score matrix
E	the residual matrix
T^2	the statistics
SPE	the statistics
X_n	the new data set
$1 - \alpha$	the confidence level
F	the F distribution
T^2_{lim}	the control limit of T^2
SPE_{lim}	the control limit of SPE
c_α	the confidence value of the standard normal distribution
N_T	the number of hidden layers of the teacher model
N_i	the number of neurons of layer i
I	the input dataset
O	the output dataset
L_t	the loss value of the teacher model
c_i	the model coefficient, i equals one to four
F_1	extraction steam flow rate [t/h]
P_{re}	the output power of renewable energy [MW]
C	the coefficient
h_s	the enthalpy value of the main steam [KJ/kg]

Q	the calorific value of the coal [KJ/kg]
W	the coefficient term
O_{pre_t}	the predicted value predicted using the teacher's model
O_r	the real value
y_1, y_2	the two variable arguments of the function MSE
$O_{m,j}$	the output of neurons j of the hidden layer m
σ_m	the activation function of the hidden layer m
In	the number of features entered for the hidden layer
$w_{i,j}$	the weight coefficient of input x_i for neuron j of hidden layer m
x_i	the input i of neurons j of the hidden layer m
b_j	the threshold of neurons j of the hidden layer m
η	the learning rate
μ	the momentum factor
N_S	the number of hidden layers of the student model
N_j	the number of neurons of hidden layer j
L	the total loss value
O_{pre_s}	the predicted value predicted using the student's model
f_1	the complex structure prediction model
f_2	the simplified structure prediction model
ts model	the model obtained based on knowledge distillation
β	the adjustable parameter of the total loss function
f_{new}	the updated model for online application
ε	the acceptable boundary value of the cumulative error range
d_{t1}, d_{t2}	the boundary values of the cumulative duration of model updates
I_{new}	the new input data
O_{new}	the new output data
$P_{turbine}$	power generation of the turbine [MW]
D	the main steam flow rate [t/h]
F_2	exhaust steam flow rate [t/h]
P_{adjust}	the adjustable power within the IES [MW]
B	the coal consumption of the boiler [t/h]
h_w	the enthalpy value of the feed water [KJ/kg]
μ	the efficiency of the boiler [%]
B', D'	B, D values after the change in renewable energy outputs [t/h]

1. Introduction

1.1. Background

The persistent emission of carbon dioxide, primarily from the use of fossil fuels, contributes to global climate change and global warming [1, 2]. Consequently, there is a consensus across various sectors of society that the transition of energy systems towards green and renewable energy sources must be accelerated [3,4]. Taking China as an example, by the end of June 2023, the installed capacity of renewable energy had historically surpassed that of thermal power units, accounting for approximately 48.80 % of the total installed capacity [5]. Against this backdrop, traditional energy production is gradually transitioning to integrated energy systems (IES) with the aim of absorbing more renewable energy, enhancing the efficiency and cleanliness of energy utilization, and reducing carbon emissions [6]. Generalized

multi-energy systems, such as multi-energy systems elaborated by Mancarella [7], the single IES described by Wu et al. [8], the regional IES introduced by Ramachandra [9], and virtual power plants detailed by Saboori et al. [10], have become hot topics in academia and industry. In this study, IES is defined as the supply side of multi-energy systems, which has higher complexity in terms of space, time, object, and operating conditions. Here, the IES possesses a stakeholder nature and pursues the maximization of asset benefits. Therefore, dispatch decision-making based on the optimization model of IES is crucial.

Many researchers have studied the complementary scheduling of multi-energy systems. These studies include considerations of uncertainty, co-optimization of multi-energy complementarity, and optimization of multiple stakeholders. For instance, Aghdam et al. [11] examined the uncertainty scheduling of electricity-hydrogen systems with virtual energy storages, and implemented operational cost optimization under the uncertainty of prices using stochastic optimization methods. Oskouei et al. [12] proposed a hybrid robust-stochastic

optimization method to address the uncertainty of electricity market prices and energy demand in systems equipped with combined heat and power generation and power-to-X storage. Mirzaei et al. [13] also used the hybrid robust-stochastic optimization techniques to address the uncertainty in multi-energy systems. Mobasser et al. [14] tackled the uncertainty optimization problem of electricity-heat-hydrogen systems and developed a model based on Hong 2m+1, information gap decision theory, and scenario-based methods. Li et al. [15] constructed a distributed collaborative optimization model for multi-energy systems based on event triggering. These studies provide scientific model support for achieving efficient and low-carbon operation in IES.

Nevertheless, the accuracy of internal component models greatly influences the implementation and execution of scheduling decisions.

1.2. Literature review

Simplified linear component models are commonly used in existing system-level optimization. For instance, Zheng et al. [16] utilized energy efficiency coefficients to depict the component models in IES, including combined heat and power generation, power-to-gas, and compressors. Similarly, Cesena and Mancarella [17] proposed a linear approximation modeling method for nonlinear network equations, and Li and Xu [18] transformed a nonlinear model into a linear one to reduce computational load. Zhang et al. [19] modeled the internal equipment in the form of an energy hub, with fixed model coefficients. While the conversion coefficient method simplifies the representation of component models and enhances model efficiency, it does sacrifice some degree of accuracy.

Apart from using simplified data models, some researchers also adopt mechanism modeling methods. Nittaya et al. [20] studied the mechanical dynamics model of post-combustion CO₂ capture equipment, and Sani et al. [21] used mechanism modeling for tank trucks and refrigeration units. Gabrielli et al. [22] proposed a modeling framework for electrochemical devices, which is based on first-principles thermodynamic dynamic modeling and develops reduced-order linear approximations. Ma et al. [23] constructed a detailed wind farm model considering free dissipation energy, resonant dissipation energy, and forced dissipation energy. Nevertheless, mechanism modeling introduces a significant level of complexity, and poses challenges to the resolution of optimization scheduling decision models.

To improve model feasibility, some researchers have expanded upon simplified data models using deep learning algorithms. Abbasimehr et al. [24] proposed a method for predicting energy demand based on convolutional neural networks. Ajbar et al. [25] constructed a predictive model for the thermal efficiency of parabolic trough collectors using artificial neural networks. Wang et al. [26] introduced a method using recurrent neural networks to predict the thermal energy utilization rate of electric thermal storage heaters. These studies demonstrate that data models based on neural networks can mine historical experiences, thereby enhancing the generalization capability of the coefficient method and reducing complexity compared to traditional mechanism models.

However, the accuracy of the model is conditional, contingent upon the consistency of the operating conditions. As renewable energy becomes increasingly integrated into the grid and component loads undergo constant adjustments, enhancing the model's adaptability to these fluctuations presents a significant challenge. Kämper et al. [27] proposed a data-driven modeling method for multi-energy systems using piece-wise linear regression. Mannsör et al. [28] highlighted the benefits of piece-wise mapping functions over models with non-piece-wise fixed cost. Benzaama et al. [29] introduced a technique to establish a thermal behavior model for residential buildings using a piece-wise affine auto-regressive genetic identification. Nonetheless, these piece-wise linear models are still offline. Further, Qin et al. [30] further proposed an adaptive piece-wise linear modeling strategy to approximate the nonlinear operating characteristics of various energy converters

adaptively. Rahbar et al. [31] improved online the efficiency of online optimization by adding constraints to make the model closer to the actual system and improving optimization methods.

Some researchers have proposed model predictive control (MPC) rolling update strategies using error feedback correction methods. For example, Labidi et al. [32], on the basis of predicting changes in power demand, implemented operational optimization for multi-energy district boilers using the MPC method. Vasilj et al. [33] constructed nonlinear thermal models and algebraic models for heating networks using nonlinear formulas and proposed a nonlinear MPC method to achieve operational optimization for multi-energy microgrids. Furthermore, Jimenez et al. [34] proposed a model-free method that uses deep reinforcement learning (DRL) to cope with the impact of external uncertainty. Ye et al. [35] put forth a real-time autonomous energy management strategy for residential multi-energy systems, leveraging the DRL methodology. Yang et al. [36] constructed a DRL-based island energy system model from the perspective of energy adaptive management, which can cope with the impact of uncertainties in renewable energy and load. Owing to its dynamic reward mechanism, DRL demonstrates adaptability to the uncertainties of external environments to a certain extent. Yet, despite the success of DRL in achieving model-free optimization, it continues to employ a fixed coefficient method for component models within IES. The research conducted by Wen and Aziz [37] provides insights into this issue. Managing multiple constraints within DRL requires further exploration. Therefore, studying the modeling and updating of component models under variable operating conditions is of practical significance.

In addition to updating methods, determining when to trigger model updates and how to evaluate the updated model pose significant challenges. Currently, the main strategies include those based on prediction accuracy thresholds, specific indicator thresholds, and real-time updates without judgment. Brauns et al. [38] proposed a method for regularly retraining and updating models for short-term vertical power flow prediction based on the accumulation of operating condition change data. Wang et al. [39] proposed a model update scheme based on parameter sensitivity analysis for rotor systems. Zhang et al. [40] discussed the update framework and methods from the perspective of multi-dimensional models, taking the machine tool cutting tool as the research object. They used physical data to update the model, minimizing the deviation between the physical model and the virtual model. Furthermore, Schreiber and Sick [41] proposed an offline updating and adaptation method for models, which improves their adaptability through model combination and fine-tuning. Huang et al. [42] adopted a direct real-time update method, using a recursive least squares algorithm with a forgetting factor to update the model in real time. Yu et al. [43] proposed a real-time update strategy based on an improved Gaussian particle filter and a Dirichlet process mixture model. The aforementioned research provides some guidance for model update strategies. Nevertheless, there is still relatively little research on the update strategies for component models, especially in the field of IES. Therefore, further research is needed to determine the optimal timing and evaluation methods.

1.3. Motivation and contributions

As revealed by the literature review, during the operation of IES, the high proportion of renewable energy integration necessitates continuous adjustment of component loads, leading to a decrease in the adaptability of component models. Although data models based on neural networks can enhance generalization capabilities and simplify complexity, they lack adaptability during operational changes and are slow to update. Therefore, this study aims to explore a method that maintains accuracy while enabling rapid online updates of the model.

Meta-learning and incremental learning in computer science offer new pathways for adaptive model updating. Meta-learning endows models with learning capabilities [44], while incremental learning

Table 1
Comparison of research status.

Ref	Component modeling method	Research level			Research object/ field	Research deficiencies
		Update time	Update method	Model adoption		
[16–18]	Coefficient-based method	×	×	×	Energy system	Insufficient model accuracy
[20–22]	Mechanism modeling method	×	×	×	Energy system	Excessive model complexity
[24–26]	Neural networks	×	×	×	Energy system	Models are offline and unable to adapt to operating conditions
[27–29]	Piece-wise linear method	×	×	×	Energy system	Models are offline and unable to adapt to operating conditions
[30]	Adaptive piece-wise linear method	✓	✓	×	Energy system	Inability to manage multi-variable coupling
[31]	Coefficient-based method	✓	✓	×	Energy system	Lagging due to reliance on deviation correction
[32–33]	MPC	✓	✓	×	Energy system	Lagging due to reliance on deviation correction
[38]	LSTM	✓	✓	×	Energy system	Frequent and delayed updates triggered by threshold-based judgments
[42]	LSTM	✓	✓	×	Machine tool feed systems	Frequent and delayed updates triggered by threshold-based judgments
[34–37]	DRL	×	✓	×	Energy system	Utilization of fixed coefficient methods for component models
[41]	Neural networks and Bayesian linear regression	×	✓	✓	Energy system	Combining multiple model complex models and not online adaptation
Proposed method	Neural networks	✓	✓	✓	Energy system	/

retains knowledge of old data through knowledge extraction [45]. Studies by Sarma et al. [46] and Wu et al. [47] confirmed the advantages of online incremental learning in energy demand forecasting. Nevertheless, when traditional neural models are used for training, they tend to exhibit high computational complexity, necessitate substantial storage resources, and pose challenges to real-time model updating.

Knowledge distillation, as a model compression technique, can transfer knowledge from complex models to simpler ones [48], and has been used in industrial processes to reduce the complexity and computational load of predictive models. For example, Li et al. [49] addressed the energy consumption problem of additive manufacturing processes by extracting hidden information or highly representative features from data related to energy consumption and reducing the complexity and computational load of prediction models based on knowledge distillation. Ai et al. [50] developed a hybrid learning framework for feature representation based on knowledge distillation and supervised contrastive learning for industrial process monitoring. However, this technique is rarely applied in energy systems. Given the dynamism and nonlinearity of IES, complex models offer high predictive accuracy but are impractical due to their extensive demands on computational

resources and time. Knowledge distillation offers a solution by transferring knowledge from complex models to simpler ones, thereby reducing computational costs while maintaining predictive accuracy.

Furthermore, while the knowledge distillation method solves the problem of how to update, the question of when to update remains challenging. Therefore, this study further explores methods for identifying operational changes. principal component analysis (PCA) is a widely used dimension reduction technique that can retain key information and has been used for industrial fault monitoring. Elsamanty et al. [51] applied PCA to the fault diagnosis of rotating machinery. Swiercz and Mroczkowska [52] studied the applicability of the PCA method in the early detection of leaks in steam boiler pipe systems in thermal power plants. This inspires us to consider using operational monitoring techniques to determine the need for model updates. In complex IES systems, PCA can effectively reduce data dimensions and simplify model complexity. By setting normal operating conditions as standard states and different conditions as anomalies, operation monitoring data can be used to detect model mismatch situations.

In summary, current component models have partially incorporated techniques such as piece-wise linear method and rolling correction to

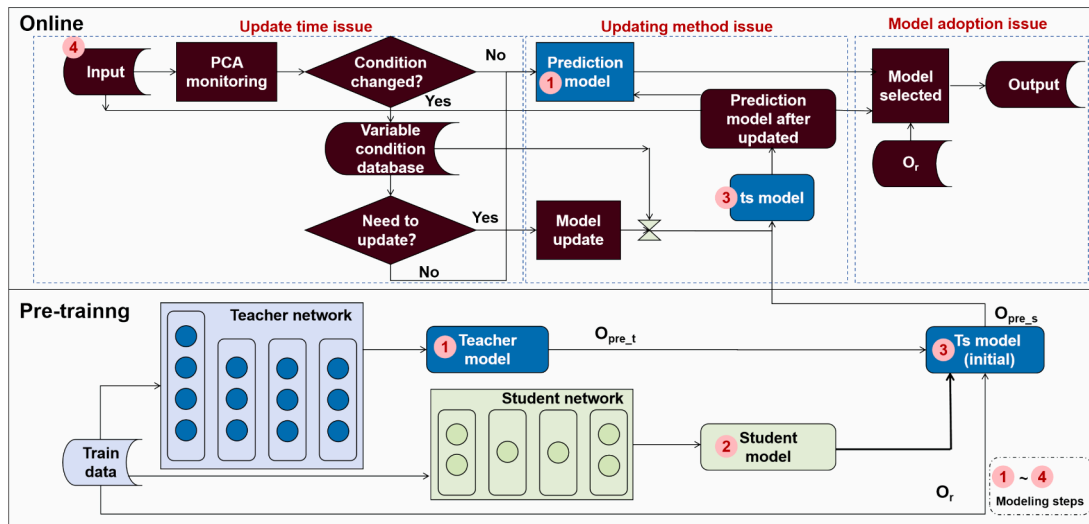


Fig. 1. Component modeling and updating method based on knowledge distillation.

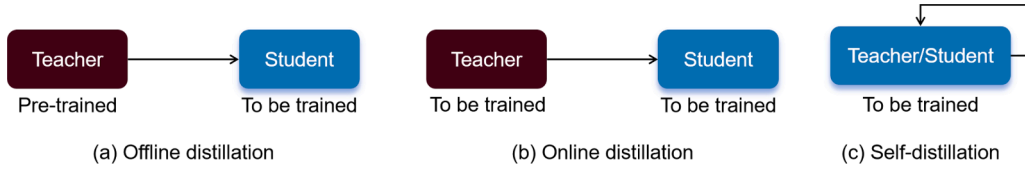


Fig. 2. Three knowledge distillation methods.

mitigate errors in variable operating conditions. Several challenges remain unresolved: (1) Update time issue: The judgment of update time is less involved in existing models. Updates are usually carried out regularly, which is lagging or triggered by thresholds, which are updated frequently, increasing the demand for computational resources. (2) Updating method issue: piece-wise linear has achieved the decomposition of operating conditions under certain conditions, but it does not consider the multi-variable coupling characteristics of operating conditions; the error correction method is only suitable when the deviation is not large; existing multi-operating condition models are still offline training. (3) Model adoption issue: Ensuring that the model after the update is better than before the update lacks scientific judgment.

Table 1 summarizes the current research status of component modeling and update methods.

This study proposes a component modeling and updating method of IES. The contributions of this study are threefold:

- (1) A component data-driven modeling and updating method of IES is proposed, integrating the model update criteria judgment, real-time model updating, and model selection.
- (2) A knowledge distillation-based method is used to achieve real-time updates of complex models, and a component condition detection model is constructed based on PCA to solve the problem of when to update the model.
- (3) Based on the quantitative analysis of renewable energy increments on the operation of boiler loads in IES, the impact of model update on the overall scheduling benefits is studied.

The rest of the paper is organized as follows. Section 2 introduces the modeling and updating methodology proposed in this work. Section 3 introduces the case settings. Section 4 introduces the obtained results and their discussion. Lastly, Section 5 contains the main conclusions and highlights recommendations for future research.

2. Methodology

Fig. 1 shows the process of the component modeling and updating method based on knowledge distillation proposed in this study. The process commences with pre-training a foundational knowledge distillation model, which encompasses a sophisticated teacher neural network and a rudimentary student neural network. The teacher model, a complex neural network model, excels in precision for stable operational contexts, as evidenced in research by Sakiewicz et al. [53], yet it faces challenges with adaptability and high computational expenses [54]. The student model, a simpler neural network model, contrasts with reduced precision but enhanced adaptability and economical resource utilization.

The predicted values obtained from the training data through the teacher model are input into the student model at the same time as the actual values, and further fine-tuning is carried out to obtain the new model (ts model) that absorbs the teacher's knowledge and distills it. The ts model has the same simple structure as the student model, but its accuracy is higher than that of the student model. During online deployment, a PCA model surveils operational conditions, capturing any deviations for subsequent integration into the ts model updates. Concurrent predictions by the updated model and the preexisting model ensue, leading to the replacement of the old component model if the

updated version maintains superior accuracy over an extended duration.

2.1. Operating condition detection method based on PCA

PCA is a multivariate process statistical monitoring technique widely used in fault monitoring in the process industry [55]. PCA reduces the dimensionality of multi-variable high-dimensional data, mapping it to a low-dimensional space and transforming it into univariate statistical processing. Both energy production factories and process industries share similarities, as they are complex processes with multi-variable coupling. This study draws inspiration from the idea of fault monitoring and applies PCA to the identification of component operating conditions as a preliminary criterion for whether the model needs to be updated.

The input data needs to be normalized.

$$\mathbf{X} = (\mathbf{X}_0 - \mathbf{X}_M) / \mathbf{X}_V \quad (1)$$

Where, $\mathbf{X}_0$ is the original data set with n observations and m variables; $\mathbf{X}_M$ is the mean of the $\mathbf{X}_0$; $\mathbf{X}_V$ is the variance of the $\mathbf{X}_0$; $\mathbf{X}$ is the normalized input data.

The covariance matrix is obtained by Eq. (2).

$$\mathbf{Z} = \mathbf{X}\mathbf{X}^T / (n - 1) \quad (2)$$

Where, $\mathbf{Z}$ is the covariance matrix.

Then, the eigenvalues and eigenvectors are calculated.

$$\mathbf{Z}\vec{\nu} = \lambda\vec{\nu} \quad (3)$$

Where, λ is the eigenvalues; $\vec{\nu}$ is the eigenvectors.

The feature values are sorted, the first k features are canceled so that their cumulative contribution is greater than d (an adjustable parameter), and the corresponding k feature vectors are selected to form the matrix $\mathbf{P}$.

$$\mathbf{X} = \mathbf{TP}^T + \mathbf{E} \quad (4)$$

Where, $\mathbf{T}$ is the score matrix; $\mathbf{E}$ is the residual matrix.

The statistics $\mathbf{T}^2$ and $\mathbf{SPE}$ are used to monitor the process conditions.

$$\mathbf{T}^2 = \mathbf{X}_n^T \mathbf{P}^T \lambda^{-1} \mathbf{P} \mathbf{X}_n \quad (5)$$

$$\mathbf{T}_{lim}^2 = k \frac{n-1}{n-k} F_\alpha(k, n-k) \quad (6)$$

Where, $\mathbf{X}_n$ is the new data set; $1 - \alpha$ is the confidence level, α is usually 0.05; F is the F distribution; $\mathbf{T}_{lim}^2$ is the control limit of $\mathbf{T}^2$.

$$\mathbf{SPE} = \mathbf{X}_n^T (1 - \mathbf{PP}^T) \mathbf{X}_n \quad (7)$$

$$\mathbf{SPE}_{lim} = \theta_1 \left[\frac{c_\alpha h_0 \sqrt{2\theta_2}}{\theta_1} + 1 + \frac{\theta_2 h_0 (h_0 - 1)}{\theta_1^2} \right]^{1/h_0} \quad (8)$$

$$\theta_s = \sum_{j=k+1}^m \lambda_j^s (s = 1, 2, 3) \quad (9)$$

$$h_0 = 1 - \frac{2\theta_1\theta_3}{3\theta_2^2} \quad (10)$$

Where, $\mathbf{SPE}_{lim}$ is the control limit of $\mathbf{SPE}$; c_α is the confidence value of the standard normal distribution.

Table 2
Comparison of three distillation methods.

Distillation method	Characteristic	Advantage	Commonly used scope
Offline distillation (This study)	Pre-training required	Flexibility in selecting large pre-trained models as teachers; Simplicity and low resource usage	Model compression
Online distillation	Simultaneous updates possible for teacher and student models	Eliminates the need for pre-training; Facilitates mutual learning among models	Knowledge fusion
Self-distillation	Capable of transferring deep-level information to shallow-level information	Eliminates the requirement for a teacher model	Model performance improvement

2.2. Component prediction model based on knowledge distillation

In order to achieve rapid online updating of complex models, this study adopts a method based on knowledge distillation. The core idea of knowledge distillation is to construct a lightweight model and train this model using the supervision information from a superior performing model, thereby obtaining an excellent model after distillation, with the aim of achieving better performance and accuracy. During online application, predictions are made using the excellent model, thereby realizing the lightweight nature of the model.

Knowledge distillation mainly includes three modes, namely offline distillation, online distillation, and self-distillation [56]. Fig. 2 shows the differences between the three knowledge distillation methods. Offline distillation generally comprises two steps: the pre-training of the teacher model and the subsequent training of the student model. The term “offline” signifies the necessity to construct a relatively complex teacher model offline, which is subsequently utilized to train the student model by extracting knowledge from it. Online distillation necessitates only a single step, wherein the teacher model and the student model are trained concurrently. Self-distillation, a unique variant of online distillation, does not differentiate between the teacher model and the student model, requiring merely a single model. In this scenario, the student learns from the knowledge it outputs.

This study adopts the offline distillation paradigm, which is the most commonly used one, and Table 2 lists the characteristics and advantages of the three methods. From the perspective of application convenience, offline distillation is the most convenient. The teacher model can include a set of models trained using different software packages, which may be located on different machines. In terms of accuracy, compared to offline distillation, online distillation can simultaneously update the teacher model and the student model, which can help improve accuracy. However, this also brings a certain computational overhead, which hinders the need for model compression. From the perspective of applicability, online distillation is more suitable for knowledge fusion or application in scenarios such as multi-modal, cross-domain [57]. Self-distillation is mainly used to enhance the performance of downstream task models, rather than for model compression or transfer learning.

Consequently, this study chooses offline distillation to achieve the purpose of model compression and improve the model update efficiency during online application. In addition, in the current background of the IES optimization operation, most component has been pre-trained with teacher models, which also supports the selection of offline distillation in this study.

2.2.1. Pre-training model construction

The pre-training process hinges on pre-training a complex neural

network and emulating it with a simpler version, yielding a model that's structurally minimalist but accuracy-comparable to the original.

Teacher model

Before constructing the teacher model, the first step is to preprocess the data, which includes handling outliers and missing values. Additionally, the selected data should be normalized to minimize the impact of features of different magnitudes on the model construction. During the construction of the teacher model, different neural network structures can be chosen to build multiple models. The most suitable structure should be selected. Assuming a complex neural network is set with N_T hidden layers, each layer having N_i neurons ($i = 1, 2, \dots, N_T$).

Given the input dataset $\mathbf{I}$ and the corresponding output dataset $\mathbf{O}$, the complex neural network is trained, including the forward computation process and the error backpropagation process. The backpropagation algorithm is the most commonly used method in neural network training. The difference between the target output and the obtained output is propagated back to the layers and adjusts the weights until the error reaches an acceptable range [58].

The objective function is defined as Eq. (11).

$$L_t = \text{MSE}(\mathbf{O}_{\text{pre}_t}, \mathbf{O}_t) \quad (11)$$

$$\text{MSE}(\mathbf{y}_1, \mathbf{y}_2) = \frac{(\mathbf{y}_1 - \mathbf{y}_2)^2}{n} \quad (12)$$

Where, L_t is the loss value of the teacher model; MSE is the average of the squared sum of the differences between the actual values of each data; $\mathbf{O}_{\text{pre}_t}$ is the predicted value predicted using the teacher's model; $\mathbf{O}_t$ is the real value; $\mathbf{y}_1, \mathbf{y}_2$ are the two variable arguments of the function MSE.

The output of neurons j of each hidden layer m is:

$$\mathbf{O}_{m,j} = \sigma_m \left(\sum_{i=1}^{In} w_{ij}x_i + b_j \right) \quad (13)$$

Where, $\mathbf{O}_{m,j}$ is the output of neuron j of the hidden layer m ; σ_m is the activation function of the hidden layer m ; In is the number of features entered for the hidden layer; w_{ij} is the weight coefficient of input x_i for neuron j of hidden layer m ; x_i is the input i of neurons j of the hidden layer m ; b_j is the threshold of neuron j of the hidden layer m .

Weight update method is as follows:

$$w_{i,j}(t+1) = w_{i,j}(t) + \eta[(1-\mu)D(t) + \mu D(t-1)] \quad (14)$$

$$D(t) = -\frac{\partial L_t}{\partial w_{i,j}(t)} \quad (15)$$

Where, η is the learning rate; μ is the momentum factor, $0 \leq \mu < 1$.

The pseudo-code of the overall neural network is Algorithm 1.

Based on Algorithm 1, the trained teacher model is obtained.

Student model

When building the student model, the structure of the model should be simplified as much as possible. An acceptable error upper limit should be set, and the simplest student model should be constructed under the condition that the error upper limit is met. Assuming the student model has N_s hidden layers (where, $N_s \leq N_T$), each with N_j neurons ($j = 1, 2, \dots, N_s$), where $N_j \leq N_i$. Given the same input dataset $\mathbf{I}$ and output dataset $\mathbf{O}$ as when training the teacher model, the initial student model is constructed based on Algorithm 1.

The loss function for student model is defined as Eq. (16).

$$L_s = \text{MSE}(\mathbf{O}_{\text{pre}_s}, \mathbf{O}_t) \quad (16)$$

Where, L_s is the loss value of the student model; $\mathbf{O}_{\text{pre}_s}$ is the predicted value predicted using the student's model.

Ts model (Model after knowledge distillation)

Using the predicted values of the teacher model and based on the overall loss function, the student model is updated to achieve knowledge distillation and obtain the ts model.

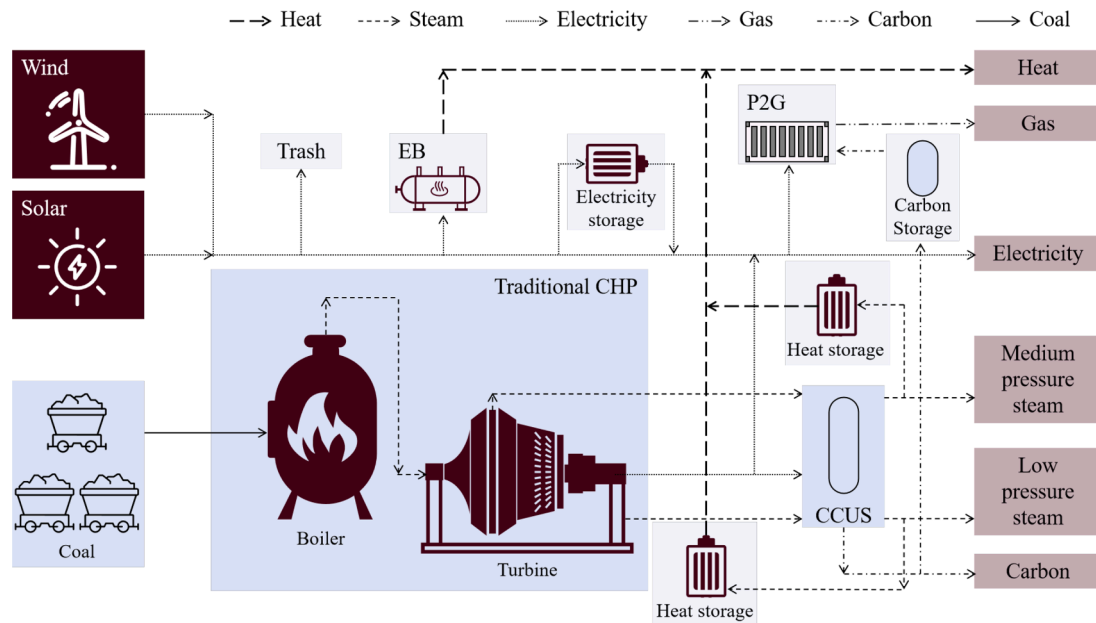


Fig. 3. The key role of coal-fired boilers in future low carbon IES.

The loss function for ts model is defined as Eq. (18).

$$\mathbf{O}_s = (1 - \beta)\mathbf{O}_{\text{pre-t}} + \beta\mathbf{O}_r \quad (17)$$

$$L = \text{MSE}(\mathbf{O}_{\text{pre-s}}, \mathbf{O}_s) \quad (18)$$

Where, L is the total loss value; O_s is the objective value for updating model; β is an adjustable parameter, namely beta parameter.

When building the ts model, the design of the beta parameter selection and the number of iterations should be considered.

For the component model of the IES, the feature data related to the component output constitutes the aforementioned input dataset **I**, and the component output is the aforementioned dataset **O**. The original prediction model before updating is usually a complex model, recorded as f_1 , that is, the teacher model; further training of the simplified structure prediction model is f_2 , that is, the student model, and the model obtained based on knowledge distillation is the ts model.

2.2.2. Online update strategy

The output model of knowledge distillation is the ts model. When applied online, the updated model f_{new} (initially the ts model) and the old model f_1 perform predictions simultaneously. If the error of the updated model prediction result over the cumulative time period is less than the error of the f_1 model or the error range is greater than the acceptable boundary value ε , an updated replacement of the model (replacing f_1 with f_{new}) is performed.

Model update trigger guidelines:

$$\left(\sum(|\mathbf{f}_l(\mathbf{I}) - \mathbf{O}_r| > |\mathbf{f}_{\text{new}}(\mathbf{I}) - \mathbf{O}_r|)\right) > d_{tl} \quad (19)$$

$$d_{l2} < \left(\sum (|f_1(\mathbf{I}) - \mathbf{O}_r| > |f_{\text{new}}(\mathbf{I}) - \mathbf{O}_r|) \right) < d_{l1}$$

$$\text{and } \sum \left(|\mathbf{f}_l(\mathbf{I}) - \mathbf{O}_r| - |\mathbf{f}_{\text{new}}(\mathbf{I}) - \mathbf{O}_r| \right) > \varepsilon \quad (20)$$

Where, ε is the acceptable boundary value of the cumulative error range; d_{i1} , d_{i2} are the boundary values of the cumulative duration of model updates, where $d_{i2} < d_{i1}$.

For continuous abnormal operating conditions, input $\mathbf{I}_{\text{new}}$ and output $\mathbf{O}_{\text{new}}$ are obtained from the abnormal operating condition library, and the ts model is fine-tuned. The fine-tuning algorithm is based on Algorithm 1, and other inputs including initial weights and thresholds are

Table 3

Component prediction model input-output.

ID	Parameter type	Parameter	Unit
1	Output	Efficiency	%
2	Input	M ain steam temperature	°C
3	Input	Main steam pressure	MPa
4	Input	Main steam flow rate	t/h
5	Input	F eed water temperature	°C
6	Input	Moisture	%
7	Input	Ash	%
8	Input	Volatile matter	%
9	Input	Net calorific value as received basis	J/kg

inherited from the ts model. The loss function selects the prediction error of the ts model alone.

3. Case introduction

Despite the ongoing adjustments in energy structure and the declining proportion, coal-fired boilers still play an indispensable role in meeting China's energy and electricity demands. This is due to the inherent variability of renewable energy sources, and the potential for an increase in coal consumption in power plants in response to growing electricity demand [59]. Achieving efficient and clean use of coal is particularly important for sustainable energy development. As shown in the Fig. 3, IES will emerge as the primary trend for enhancing the efficiency of traditional coal-fired boilers [60]. Within the IES, boilers are responsible for converting fossil energy into steam for further energy conversion and utilization. Rational adjustment of boiler load, reduction of energy consumption, enhancement of renewable energy absorption rate, and subtle energy transition are key to the energy transformation of coal-fired power plants. Given the existing quantity and significance of coal-fired boilers, this study uses the coal-fired boiler in IES as an example to validate the proposed component modeling and updating method. The data is derived from the actual operation data of a 220.00 t/h high-temperature ultra-high-pressure circulating fluidized bed boiler in Zhejiang Province.

Specifically, the historical data selected spans from March 10, 2023, to May 20, 2023, with a frequency of one minute and a total of 86,793 valid data points. The efficiency of the boiler is related to the following

Table 4
Supplementary parameters for boiler operating conditions.

ID	Parameter	Unit
1	Furnace temperature	°C
2	Flue gas oxygen concentration	%
3	Primary and secondary air ratio	/
4	Primary air volume	Nm ³ /h
5	Secondary air volume	Nm ³ /h
6	Coal feeder frequency	Hz
7	Induced draft fan frequency	Hz
8	Forced-draft blower	Hz

Table 5
Training and condition testing dataset.

Case	Date range	Description
Train	2023/3/10–2023/4/30	Basic modeling conditions
Test: case #0	2023/5/4–2023/5/11	Case with partial efficiency variations
Test: case #1	2022/2/16–2022/2/25	Case with incorrect calorific value unit
Test: case #2	2023/3/27–2023/4/1	Case with part of data extracted from training data
Test: case #3	2022/2/16–2022/2/25	Case with calorific value unit modified based on case #1
Test: case #4	2022/4/20–2022/4/29	Case with changed operating conditions

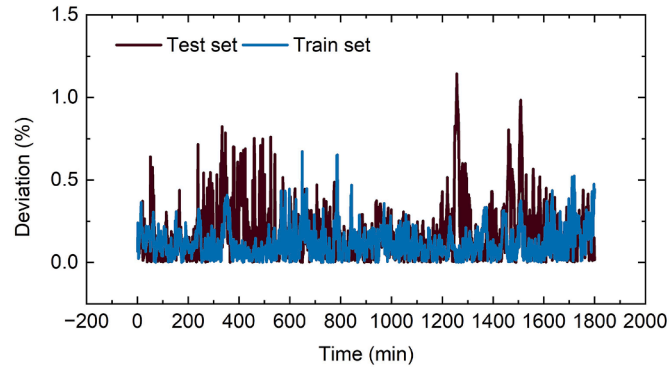


Fig. 4. Results of baseline component prediction model.

factors: main steam temperature, main steam pressure, main steam flow rate, feed water temperature, and the properties of coal (including moisture content, ash content, volatile matter, and net calorific value on a received basis). The inputs and outputs of the boiler efficiency prediction model are shown in Table 3.

In addition to the boiler efficiency-related inputs and outputs in Table 3, the parameters in Table 4 are further added as monitoring parameters for boiler operating conditions. Table 5 shows the description of the dataset used in this study.

To further substantiate the universality of the proposed component modeling and updating method, operational data of a steam turbine has been chosen for additional validation. Utilizing the inflow, outflow, and extraction flow rates, as well as the temperature and pressure information of the steam turbine, a predictive model for the power generation of the steam turbine has been constructed. The hardware environment is an Apple M1 with 8 GB RAM, and the operating system is macOS.

4. Results and discussions

4.1. Construction of the baseline component prediction model

A five-layer neural network is used as the component basic model,

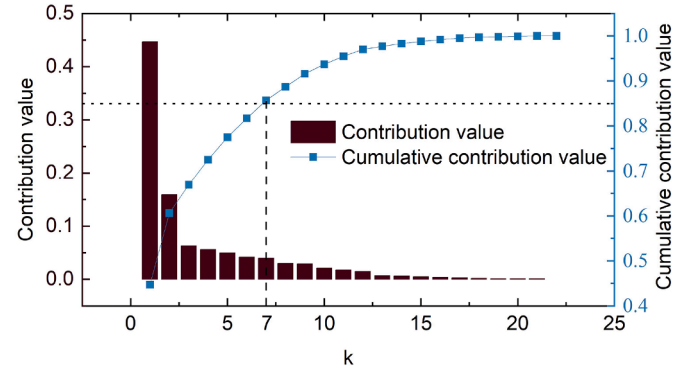


Fig. 5. Principal components contribution.

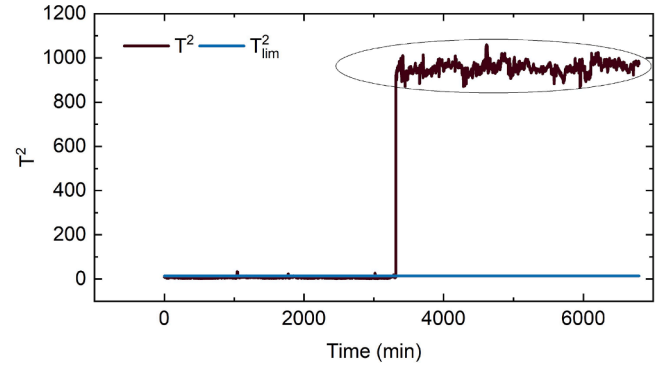


Fig. 6. Condition detection results - T^2 of case #0.

with the number of neurons in each layer being 500, 500, 500, 500, and 1, respectively. The activation function is “relu” for all layers. Optimizer selects “adam” optimizer, and the training data accounted for a random 70 % of the total data. The prediction results of the model are shown in Fig. 4. The y-axis “Deviation” in Fig. 4 represents the percentage of the absolute deviation between the predicted and actual values relative to the actual values. The mean squared error (MSE) on the test set is 0.14 %. The mean absolute percentage error on the training set is 0.12 %, with a maximum of less than 0.67 %. For the test set, the mean absolute percentage error is slightly higher at 0.18 %, but still maintains a maximum of less than 1.14 %. These results indicate that the prediction accuracy of the basic model met the general requirement of industrial prediction (5 %).

4.2. Validation of the modeling and updating method

The modeling and updating method proposed in this study

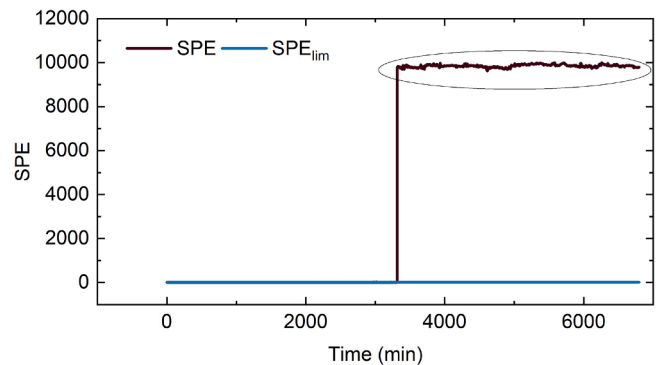


Fig. 7. Condition detection results - SPE of case #0.

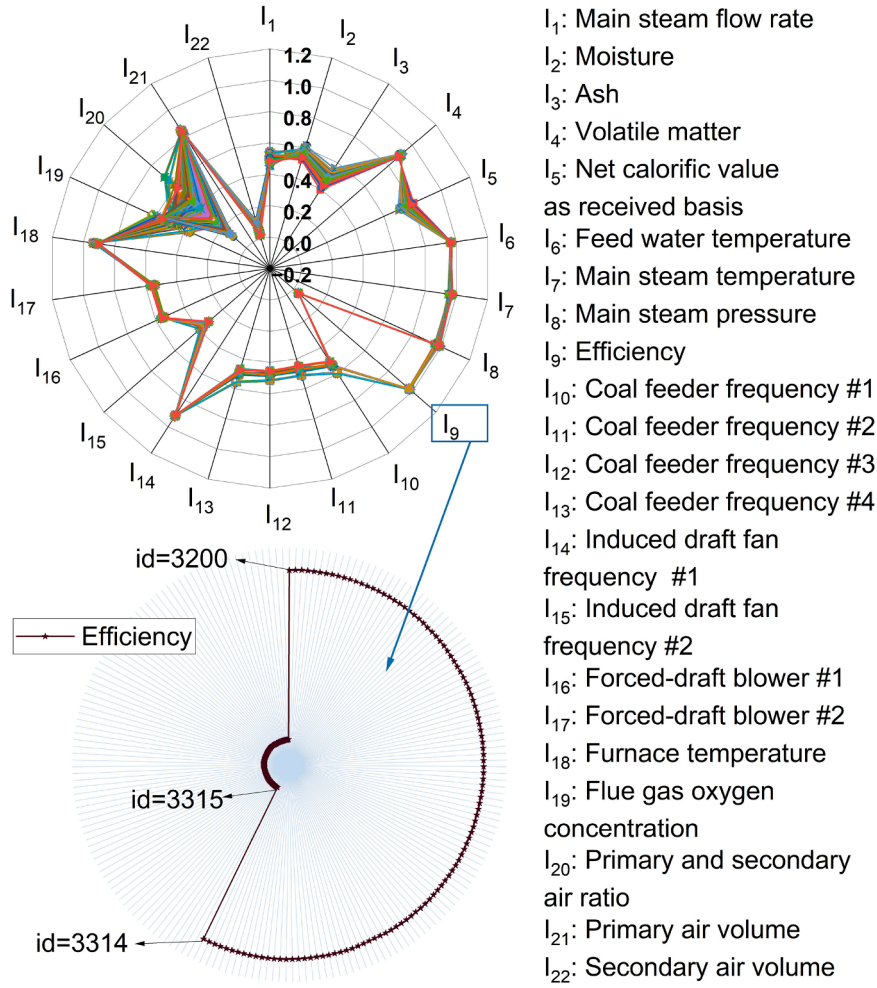


Fig. 8. Actual distribution of conditions in the case #0.

encompasses three components: triggering of model updates, the update process itself, and the adoption of the updated model. Initially, the effectiveness of PCA and the knowledge distillation model is validated. Subsequently, online applications and analytical discussions are further conducted based on this foundation. Finally, another component from the IES, the steam turbine, is selected to validate the universality of the model.

4.2.1. Effectiveness of operating condition detection

A confidence level of 95.00 % is chosen, corresponding to a value of 0.05 for α . The training data period is the same as in Section 4.1, and the test data is from 13:36 on May 4, 2023 to 14:37 on May 11, 2023. According to the cumulative contribution rate calculation results, when the number of principal components is seven, the cumulative contribution rate reaches 85.00 %, as shown in Fig. 5.

Figs. 6 and 7 show the operating condition detection results of the test data (case #0). Among them, the thresholds of T^2 and SPE are 14.07 and 8.07, respectively. As can be seen from Figs. 6 and 7, the proposed PCA method detected changes in the operating conditions of test data starting from the 3315th point. Operating conditions near 3315th point are extracted from the data and plotted.

As shown in Fig. 8, starting from 3315th point, there is a significant decrease in the efficiency data of the boiler. This further leads to changes in T^2 and SPE. Specifically, the upper half of Fig. 8 presents specific values of operating conditions at each time point, including main steam flow rate, moisture, ash, etc. Each variable has been normalized, hence the distance from the center to the boundary of the circle represents the

value range from -0.2 to 1.2, with each polygon representing an operating condition time point. It can be observed that the edges of most polygons overlap. Among them, the parameters of flue gas oxygen concentration and primary-secondary air ratio show edge shapes, indicating changes in the operating conditions. The edge of boiler efficiency has a step, where one value jumps to a position close to 0. Further, boiler efficiency data between points 3200 and 3398 has been extracted and plotted in the lower half of Fig. 8. It can be seen that the boiler efficiency indeed experiences a sudden change to near 0 during the time points 3315 to 3398, beginning at the 3315th point.

In addition, the effectiveness of the operating condition detection method proposed in this study is further tested on cases #1-#3. As shown in Fig. 9, in case #1, each data point is identified as a change in operating conditions due to a significant deviation in heat value. For case #2, the proposed method successfully detects certain changes in operating conditions. The anomaly identified in case #3 do indeed deviate from the training conditions. These findings demonstrate the efficacy of the method proposed in this study for timely monitoring of changes in operating conditions.

4.2.2. Parameter selection and modeling of knowledge distillation

The component prediction model built in Section 4.1 is selected as the teacher model. Furthermore, a simplified student model is constructed. The student model maintains the neural network structure of the five hidden layers in the teacher model, but reduces the number of units in each hidden layer to 10, 10, 10, and 1 respectively. All other parameters remain unchanged. After training, an MSE of 0.30 % is

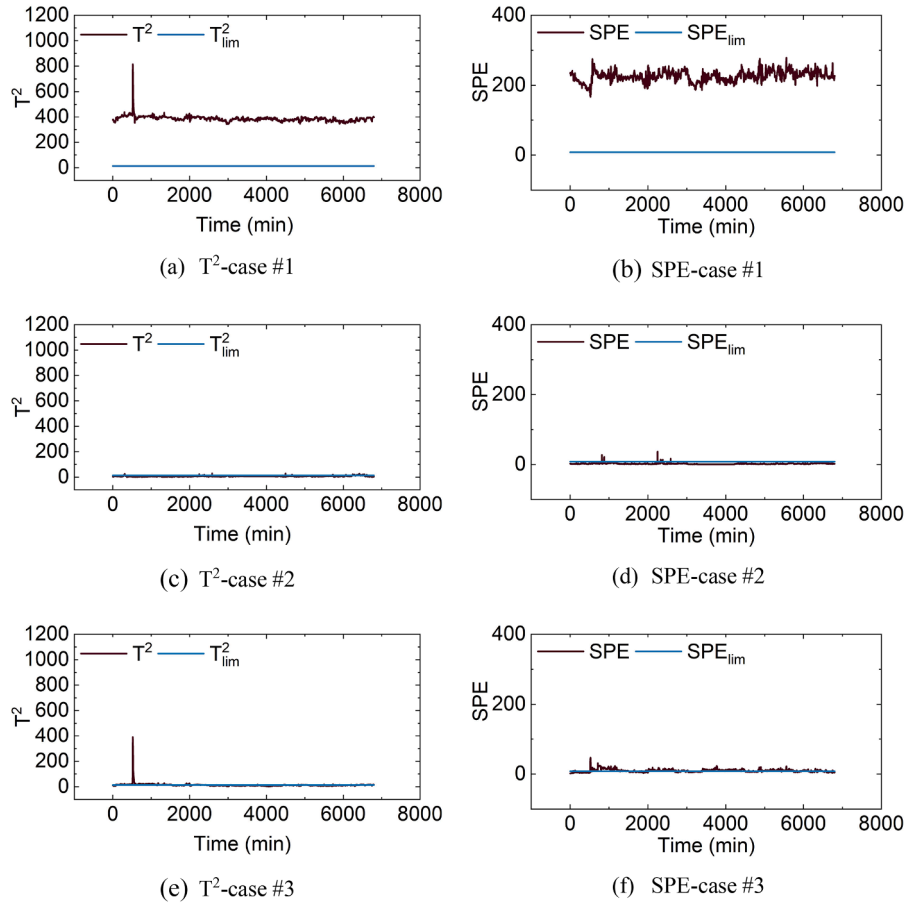


Fig. 9. Test results of condition detection for the other three test sets.

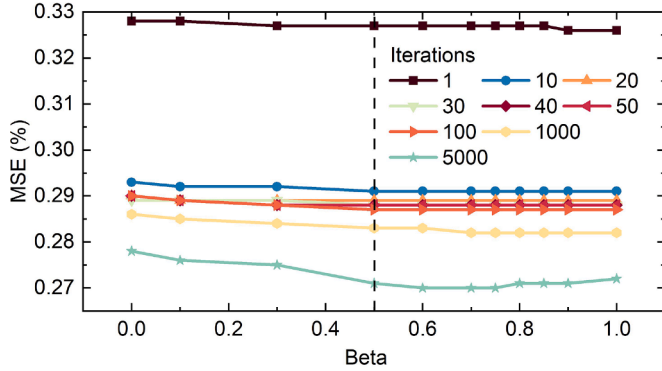


Fig. 10. Parameter selection of knowledge distillation model – beta coefficients.

obtained, which is higher than the basic model's MSE.

Upon completion of the student model construction, a knowledge distillation operation is conducted. As per Eq. (17), it is discerned that the effect of knowledge distillation is influenced by the beta parameter. As depicted in Fig. 10, tests are performed with varying beta coefficients to investigate their impact on knowledge distillation. The results indicate that when the number of iterations is less than 100, the model error curve exhibits a declining trend as beta increases from 0 to 0.50. However, when beta is greater than 0.50, the MSE is less affected by the increase in beta. From the perspective of the number of iterations, there is a significant improvement in MSE after increasing from 1 to 10 iterations. When the number of iterations is between 10 and 1000, the MSE curves are relatively close, with an MSE range between 0.28 % and 0.30

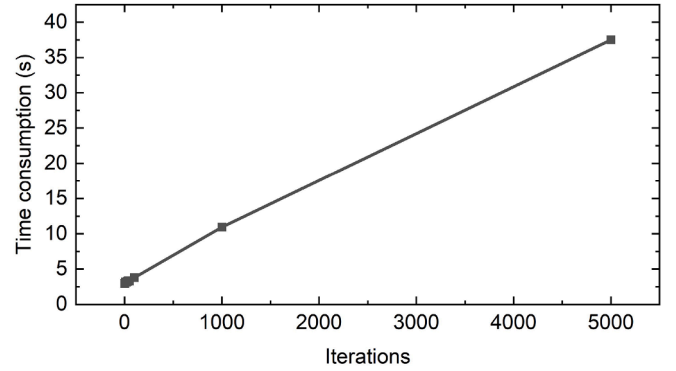


Fig. 11. Parameter selection of knowledge distillation model – iterations.

%.

Fig. 11 shows the change in time consumption under different knowledge distillation iteration numbers when beta is chosen to be 0.50. As the number of iterations increases, the time consumed also continues to increase. The time consumption is relatively close when the number of iterations is between 1 and 100, but there is a significant increase in time consumption when the number of iterations reaches 1000.

Considering both the improvement effect of model MSE and time consumption, this study chooses a beta of 0.50 and 10 iterations for knowledge distillation to train the initialized ts model.

Based on case #3, Fig. 12 shows the prediction deviations of the initial model of knowledge distillation. It is observed that the performance of the model after knowledge distillation exhibits a certain degree of enhancement compared to the original student model, with the

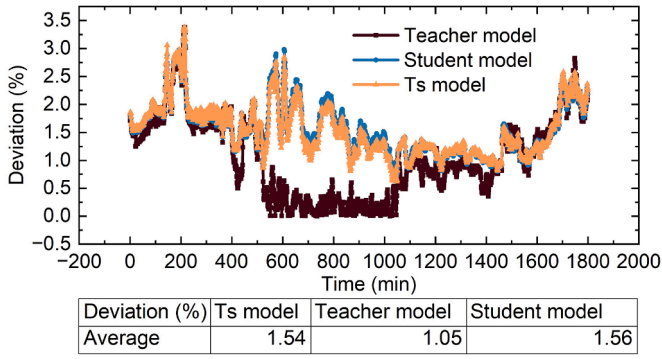


Fig. 12. Results of constructing the initial model of knowledge distillation.

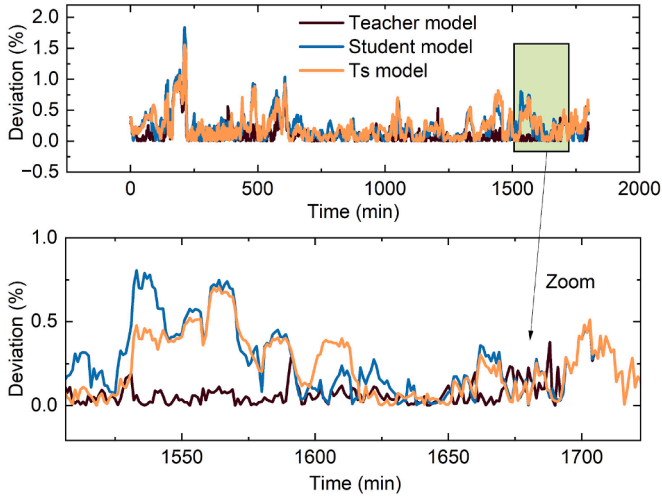


Fig. 13. Online model update application effect - based on case #3.

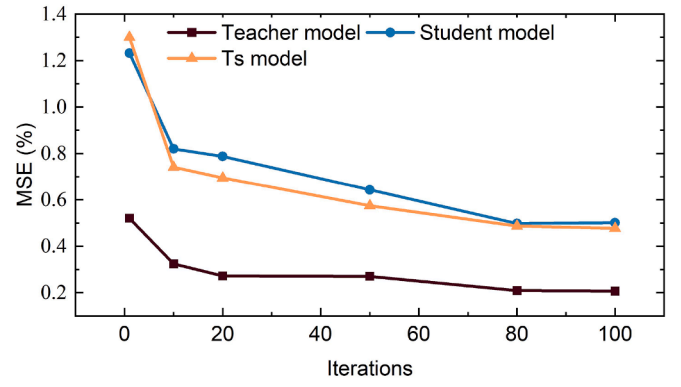
average deviation decreasing by 1.39 %. This suggests that knowledge distillation can absorb the experience of the teacher model while simplifying the model, thereby improving the accuracy of the model.

4.2.3. Online application based on knowledge distillation model

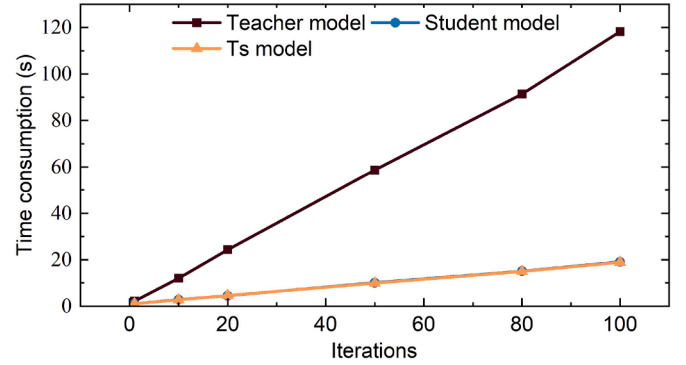
The model distilled from knowledge as generated in Section 4.2.2 is applied on different operation cases, with ten iterations of model update computations conducted and their average taken. Case #3 is utilized as the abnormal operating condition data. When the model requires updating, Fig. 13 demonstrates the iterative update effects based on three models, including the teacher model, the student model, and the ts model.

Compared to Fig. 12, the average deviation of ts model is 0.23 %, which is 1.31 % lower than the 1.54 % before the online update. In terms of time consumption, the online update of the ts model requires only 2.77 s for ten iterations, while the student model and teacher model take 2.81 s and 12.03 s respectively. The results indicate that the model update based on the knowledge distillation model can significantly reduce the online model update time. Compared with the teacher model, the time consumption is reduced by 76.67 %. Compared with the student model, the time consumption is shortened by 1.43 %. In addition, the model deviation based on knowledge distillation is reduced by 9.66 %.

Fig. 14 compares the results of online updates based on different models. The results show that as the number of iterations increases, the MSE decreases and the slope decreases. However, the time consumption is linearly positively correlated with the number of iterations. The MSE of the model after knowledge distillation on this case #3 is less than that of the student model under the same number of iterations, and the time consumption of the model after knowledge distillation is significantly



(a) MSE



(b) Time consumption

Fig. 14. Results of online updates based on different models - case #3.

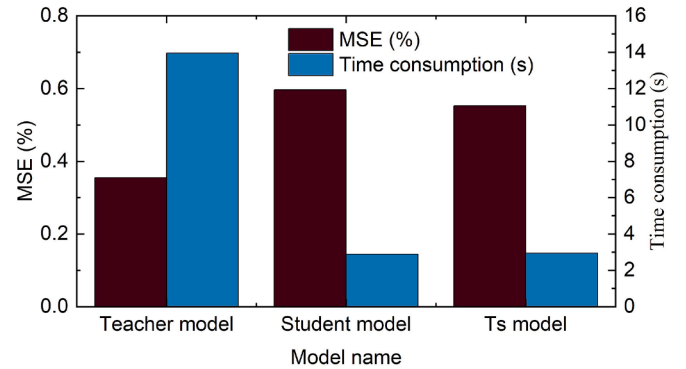


Fig. 15. Results of online updates based on different models - case #4.

reduced compared to the teacher model.

Fig. 15 illustrates the impact of online model updates under the changing operating conditions represented by case #4. The results reveal that the trend observed in case #3 is consistent with the results validated on case #4. The model after knowledge distillation continues to exhibit advantages in terms of MSE and time consumption. The MSE following the online update computations based on the ts model is lower compared to the update directly based on the student model. From the perspective of computational efficiency, it also significantly surpasses the update based on the teacher model.

4.2.4. Universality of the method

As illustrated in Table 6, the model proposed in this study is equally applicable to the updates of the turbine model. The update strategy based on the method of this study yields smaller MSE under different

Table 6
Updating results of the turbine model using proposed method.

Method	Iterations		100		1000		2000	
	10	Time (s)	MSE (%)	Time (s)	MSE (%)	Time (s)	MSE (%)	Time (s)
Ts model	0.053	4.61	0.029	4.85	0.0086	9.26	0.0054	14.10
Student model	0.056	5.13	0.043	5.76	0.020	19.03	0.019	46.43
Teacher model	2.37	31.96	0.0074	253.51	0.0023	13,541.89	0.0035	4736.52

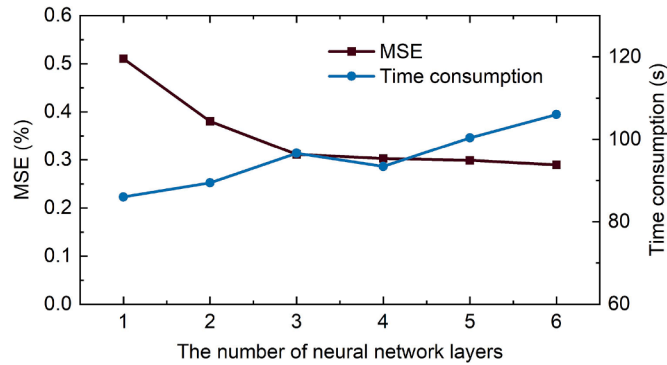


Fig. 16. The influence of neural network layers on prediction performance.

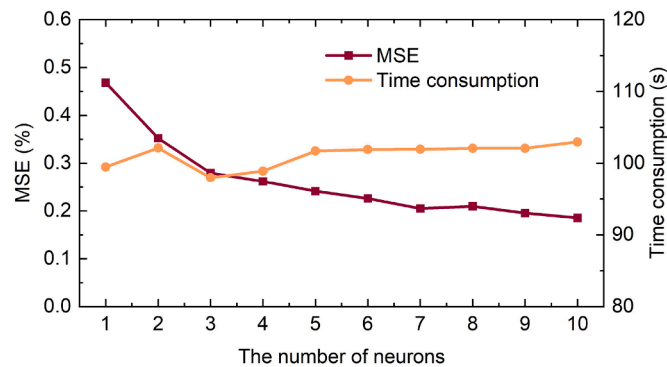


Fig. 17. The influence of the number of neurons on prediction performance.

iteration counts compared to direct updates based on the student model, with an average reduction of 41.42 %. Furthermore, the effect of MSE reduction becomes more pronounced as the number of iterations increases. Compared to direct model updates based on the teacher model, the update unfolded based on the ts model proposed by this study significantly reduces computational time. When the number of iterations is 10, the turbine model update of this study reduces the time by 27.35 s, compared to direct updates on the teacher model. For up to 2000 iterations, the computational time can be reduced by 4722.42 s. It is evident that the model update method based on knowledge distillation proposed in this study has effective advantages in improving computational efficiency.

4.3. Impact of model structure on prediction effect

To further explore the impact of different structures on knowledge distillation, this study tests the impact of different neural network layers. Keeping the teacher model constant, as shown in Fig. 16, it is observed that as the number of layers in the student model increases, the model MSE gradually decreases. In contrast, the time consumption shows a gradual increasing trend. However, it is noteworthy that as the number of neural network layers increases, its impact on model MSE will gradually stabilize. Therefore, after considering factors including model MSE and time consumption, it can be inferred that an optimal number of

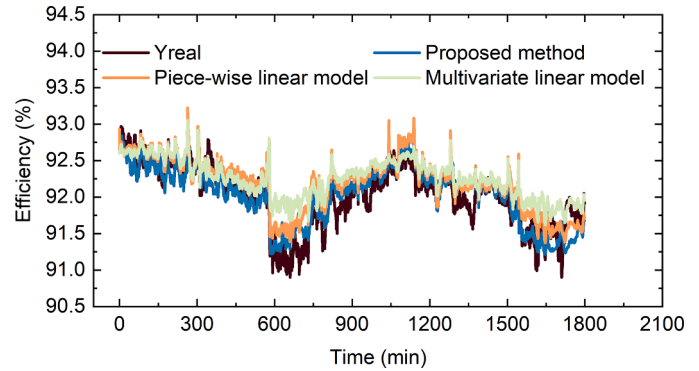


Fig. 18. Prediction results compared to traditional methods - on training set.

neural network layers exists. This optimal layer count can ensure model performance while considering computational efficiency.

Fig. 17 illustrates the impact of the number of neurons on the effectiveness of knowledge distillation. The findings suggest that under a constant number of neural network layers, as the number of neurons increases, the time consumption experiences slight oscillations but does not significantly increase. The increment in time consumption is less than the impact caused by the number of neural network layers. Moreover, under the same number of neural network layers, as the number of neurons increases, the MSE shows a declining trend and gradually stabilizes.

4.4. Adaptability analysis: comparison with traditional methods

In order to further elucidate the adaptability of the method proposed in this study, a comparison has been made with two of the most commonly used variable condition models, namely the multivariate linear model and the piece-wise linear model. The case #3 is selected as test data.

First, a multivariate linear model is constructed. The calculation time is 1.29 s, and the model score is 0.76. Second, a five-segment linear model is constructed based on the main steam flow as the segmentation basis. Fig. 18 illustrates the prediction effect of the three methods on the training set. “Yreal” is the true value of boiler efficiency. It can be seen that on the training set, the prediction value of the proposed method is closest to the original value, followed by the piece-wise linearization method, and the bias of multivariate linear prediction is the largest.

Furthermore, the variable condition prediction test is carried out based on three methods, as shown in Table 7. From case #3, the deviation of the piece-wise linearization result is more significant than that of the direct linearization result. It can be seen that the average deviation obtained by the neural network method is 12.61 % lower than that of the piece-wise linear method, and 3.49 % lower than the multivariate linear method. The results of the two training sets are similar, but when extrapolated to changing operating conditions, the generalization ability of piece-wise linearization is worse. The main reason is that piece-wise linearization is challenging to consider multiple variable coupling. The method proposed in this study has better generalization ability than traditional multiple linear and piece-wise linear methods, and its prediction is close to that of the basic model, with a more

Table 7
Comparison of condition adaptability with traditional methods.

Method	Case #3 Average deviation (%)	Maximum deviation (%)	MSE (%)	Case #4 Average deviation (%)	Maximum deviation (%)	MSE (%)	Train Average deviation (%)	Maximum deviation (%)	MSE (%)
Multivariate linear methods	1.60	3.37	8.80	1.46	3.42	7.88	0.27	1.25	0.43
Piece-wise linear methods	1.76	3.67	11.21	1.47	3.53	7.97	0.23	0.96	0.27
The proposed method	1.54	3.37	8.35	1.04	3.26	4.62	0.18	0.98	0.17

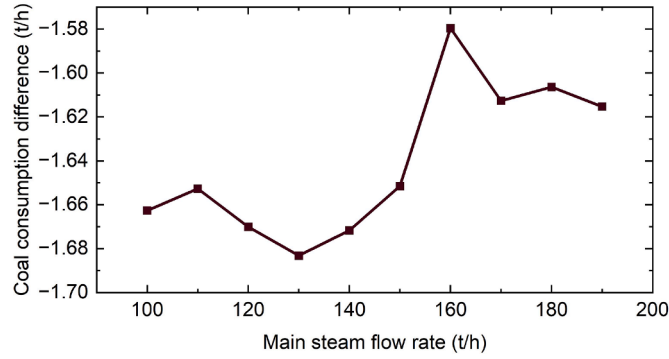


Fig. 19. Impact of each MW increase in renewable energy on boiler coal consumption.

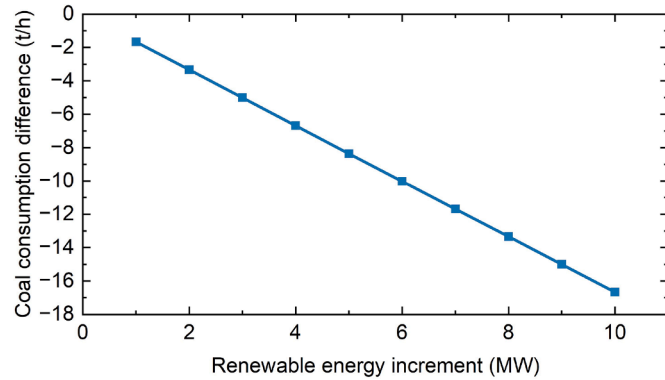


Fig. 20. Impact of renewable energy increment on coal consumption (Main steam flow rate=100.00 t/h).

straightforward structure and faster update speed.

4.5. Necessity analysis: impact of component model updates on IES scheduling

The fluctuation in the proportion of renewable energy affects the load of the component of IES. The volatility of component load further poses higher challenges to the accuracy of the component model. In order to delve into the impact of model updates on overall scheduling under the condition of high proportion of renewable energy access, this study first conducts a sensitivity analysis of renewable energy to component load (formula derivation can be found in [Appendix A](#)), and then analyzes the impact of component model errors on overall scheduling.

As depicted in [Fig. 19](#), when the flexible control of energy storage is disregarded, an increase of 1.00 MW in the power generation from renewable energy sources results in a slightly varied reduction in coal consumption under different main steam flow rates, with an average decrease of 1.64 t/h. [Fig. 20](#) demonstrates that, given a constant main

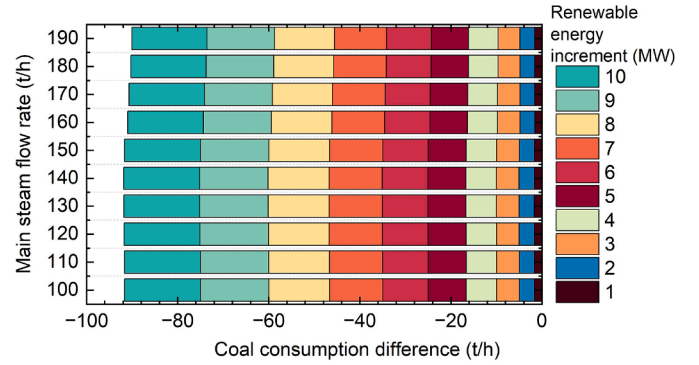


Fig. 21. Impact of renewable energy increment on coal consumption (under different main steam flow rates).

Table 8
The reflection of MSE on boiler efficiency.

Id	MSE (%)	RMSE	Efficiency (%) (Reverse normalization)
1	0.01	0.01	0.049
2	0.10	0.032	0.16
3	1.00	0.10	0.50
4	5.00	0.22	1.11
5	10.00	0.32	1.57

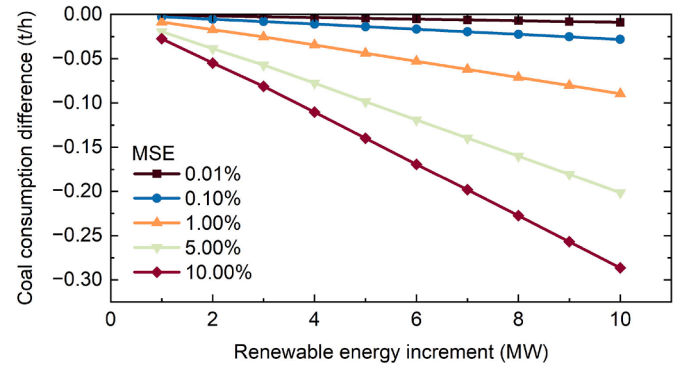


Fig. 22. The impact of model MSE on coal consumption.

steam flow rate, the augmentation in renewable energy and the corresponding decrease in coal consumption exhibit a linear relationship.

Further, [Fig. 21](#) demonstrates that the decrease in coal consumption resulting from a unit increase in renewable energy varies under different main steam flow rates. With a constant increase in renewable energy, a larger main steam flow rate corresponds to a smaller reduction in coal consumption, indicating higher boiler efficiency under high load conditions. Specifically, a unit increase in renewable energy can reduce coal consumption by 1.62 t/h at a main steam flow rate of 190.00 t/h, and by 1.65 t/h at a rate of 100.00 t/h. It is also observed that a smaller main

Table 9

Boiler efficiency deviation on coal consumption and operational profits.

Item	Boiler Efficiency (%)				
	89.50 (MSE=1.00%)	89.00	88.00	87.00	86.00
Coal consumption (t/h)	23.44	23.57	23.84	24.11	24.39
Coal consumption difference (t/h)	0.13	0.26	0.53	0.80	1.08
Profit difference (¥/day)	2950.95	5971.08	12,077.86	18,325.02	24,717.47

Table 10

Boiler efficiency deviation on power and operational profits.

Item	Boiler Efficiency (%)				
	89.50 (MSE=1.00 %)	89.00	88.00	87.00	86.00
Main steam flow rate (t/h)	186.45	185.40	183.32	181.24	179.16
Power difference (MW)	0.075	0.15	0.30	0.46	0.61
Profit difference (¥/day)	1084.64	2182.38	4364.76	6547.14	8729.51

Algorithm 1

Construction of teacher model.

Inputs: Dataset **I** and corresponding output dataset **O**, N_H hidden layers, layer neuron settings N_i , learning rate η , momentum factor μ , activation function settings.

Outputs: Trained neural network model parameters (weights and thresholds of each neural network layer), training error.

- (1) Network initialization, randomly generate weights and thresholds;
- (2) Input training dataset **I**, **O**;
- (3) Based on input **I**, calculate the output of each hidden layer neuron and the output of the output layer **O'** according to Eq. (13);
- (4) Calculate the error between the hidden layer and the output layer;
- (5) Update connection weights and thresholds;
- (6) Calculate the error function to determine whether it is less than the given learning threshold. If satisfied, the learning ends and jumps to step 7), otherwise jumps to step 2);
- (7) Retain weights and thresholds, output training error.

steam flow rate does not necessarily lead to higher coal consumption for the same increase in renewable energy output. These findings suggest that integrating varying proportions of renewable energy can have diverse effects on boiler load.

In order to further analyze the impact of model errors on IES scheduling decisions and to highlight the importance of model adaptive updates, tests are conducted under different MSE.

As shown in Table 8, it reflects the MSE in terms of boiler efficiency. A 1.00 % MSE, after reverse denormalization calculation, will result in a 0.50 % deviation in boiler efficiency.

As shown in Fig. 22, when the MSE is 0.01 %, 0.10 %, 1.00 %, 5.00 %, and 10.00 %, respectively, it will produce an average coal consumption deviation of 0.0048 t/h, 0.015 t/h, 0.048 t/h, 0.11 t/h and 0.16 t/h. Calculated from the price of thermal coal at 950 ¥/t, a 1.00 % MSE should result in a profit difference of ¥ 1094.40 per day.

Furthermore, based on a fixed power generation (i.e., fixed main steam flow rate), the impact of model deviation on coal consumption is calculated. Assuming the main steam flow is 190.00 t/h and using 90.00 % as the baseline boiler efficiency, Table 9 presents the influence of changes in boiler efficiency within 4.00 % on coal consumption and profit.

Some markets have set clear deviation assessment methods for auxiliary service market entities. For example, the Zhejiang market imposes a double assessment fee penalty on winning entities that cannot provide auxiliary services. Additionally, an analysis based on fixed coal consumption is supplemented to calculate the impact of model deviation on power generation. Assuming the boiler coal consumption is 23.00 t/h, also with 90.00 % boiler efficiency as the baseline efficiency, Table 10

shows the impact of boiler efficiency changes within 4.00 % on power generation and cost. As a result, a 1.00 % MSE caused by the model not being updated in time may result in a power deviation of 0.075 MW. Calculated at an electricity price of 300 ¥/MWh, this could result in a penalty cost of ¥ 1084.64 per day.

Therefore, it can be concluded that for IES, the adaptive update of the model is of paramount importance.

In summary, the component modeling and updating method based on knowledge distillation proposed in this study has more significant advantages in terms of computational efficiency and adaptability to operating conditions than traditional methods and can capture changes in the operating conditions of IES.

5. Conclusions

Making decisions based on models is currently a fundamental way for IES to manage energy and control. This study proposes a data-driven modeling and updating method to address the problem of reduced adaptability of component models in IES under changing operating conditions. Taking boiler efficiency as an example, the proposed method can monitor changes in operating conditions and carry out timely model updates. Regarding real-time performance, the knowledge distillation model reduces the time consumption by 76.67 % compared to direct updates based on complex models. In terms of adaptability to operating conditions, compared with traditional multivariate linear and piecewise linear methods, the proposed method has better generalization ability on the test set, with an average deviation of 12.61 % lower than piecewise linear and 3.49 % lower than multivariate linear method. From the perspective of the necessity of model updates, an increase of 1.00 MW in renewable energy can reduce coal consumption by 1.64 t/h, and a 1.00 % MSE caused by untimely model updates may lead to a power deviation of 0.075 MW. This finding further emphasizes the importance of model updates. Furthermore, further analysis shows that the increase in neural network layers has a more noticeable impact on errors after knowledge distillation.

The study's contribution and critical findings can be listed as follows:

- (1) A component modeling and updating method of IES is proposed, integrating the judgment of model update criteria, real-time model updating, and model selection. The results show that the method is real-time, adaptable, and necessary.
- (2) A knowledge distillation-based method is employed, supervising existing complex component models to construct a lightweight model, offering a novel strategy for real-time model updating. Concurrently, a PCA-based component condition detection method is utilized, presenting a new solution for determining the appropriate time to update data models.
- (3) An analysis is conducted, starting from the growth of renewable energy, deducing changes in boiler coal consumption, and examining whether the update of the boiler model influences the overall benefits of scheduling. This provides a novel perspective on the impact of renewable energy growth on boiler operation and the significance of model updating in scheduling efficiency.

Future research will be dedicated to reducing the dependence on the quality and quantity of data, integrating mechanism-based models with data-driven methods to conduct research on the fusion of mechanisms

and data, and exploring modeling strategies in small sample scenarios. Furthermore, improvements to the PCA condition detection method will be pursued, along with the exploration of strategies to enhance monitoring efficiency under different operating conditions.

CRedit authorship contribution statement

Xueru Lin: Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Wei Zhong:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Xiaojie Lin:** Writing – review & editing, Supervision, Methodology. **Yi Zhou:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Long Jiang:** Conceptualization, Methodology, Supervision. **Liuliu Du-Ikonen:** Conceptualization, Methodology, Supervision. **Long Huang:** Conceptualization, Methodology, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A

A sensitivity analysis of renewable energy to the boiler load

Initially, a turbine model is constructed, with inputs including steam inflow, extraction steam flow, and exhaust steam flow, and the output being power generation, forming a multivariate linear regression equation.

$$P_{\text{turbine}} = c_1 \times D + c_2 \times F_1 + c_3 \times F_2 + c_4 \quad (\text{A.1})$$

Where, P_{turbine} is power generation of the turbine; c_i is the model coefficient, i equals one to four; D is the main steam flow rate; F_1 is extraction steam flow rate; F_2 is exhaust steam flow rate.

According to Eq. (A.2), the power balance formula of the IES, it is found that without considering the flexible scheduling space brought by storage and other component, for every additional 1 MW of renewable energy, the steam production of the boiler needs to decrease by 13.74 t/h.

$$P_{\text{turbine}} + P_{\text{re}} + P_{\text{adjust}} = P_{\text{demand}} \quad (\text{A.2})$$

$$D = (P_{\text{demand}} - P_{\text{re}} - P_{\text{adjust}} - c_4 - c_2 \times F_1 - c_3 \times F_2) / c_1 \\ = -P_{\text{re}} / c_1 + C \quad (\text{A.3})$$

Where, P_{re} is the output power of renewable energy; P_{adjust} is the adjustable power within the IES; P_{demand} is the total demand power; C is the coefficient part converted according to Eq. (A.1) and (A.2).

Combining with Eq. (A.4), the calculation formula for the quantity of coal is obtained. Through this formula, the relationship between the increase in renewable energy and the decrease in the quantity of coal can be derived. (Eq. (A.5) and (A.6))

$$B = D(h_s - h_w) / \mu Q = D \times W / \mu \quad (\text{A.4})$$

$$\begin{aligned} B' - B &= D' \times W / \mu' - D \times W / \mu \\ &= D' \times W / F(D') - D \times W / F(D) \end{aligned} \quad (\text{A.5})$$

$$D' = D - 1 / c_1 \quad (\text{A.6})$$

Where, B is the coal consumption of the boiler; h_s is the enthalpy value of the main steam; h_w is the enthalpy value of the feed water; Q is the calorific value of the coal; μ is the efficiency of the boiler, which is a function of D , denoted by F ; W is the coefficient term; B' , D' are B , D values after the change in renewable energy outputs.

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